Process scheduling examples:

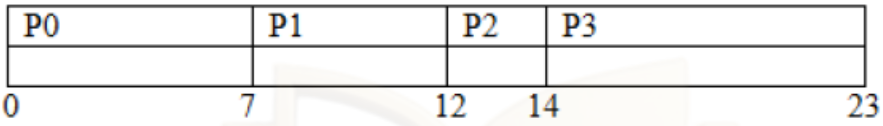
Q1: Find the average waiting time and average turnaround time for executing the following processes using FCFS (first-come first-service) scheduling?

Process	Burst time
P0	7
P1	5
P2	2
P3	9

Answer:

The first step of the solution is founding the Gantt chart.

Gantt chart:



The waiting time of P0 = 0
The waiting time of P1 = 7
The waiting time of P2 = 12
The waiting time of P3 = 14
The average waiting time = (0 + 7 + 12 + 14) /4 = 33/4 = 8.25

The turnaround time of p0 = 7
The turnaround time of p1 = 12
The turnaround time of p2 = 14
The turnaround time of p3 = 23
The average turnaround time = (7+12+14+23)/4 = 56/4 = 14

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Q2: Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process using (1) Preemptive short-job first (2) Non-preemptive short-job first?

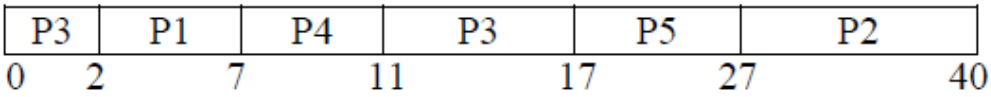
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Process	P1	P2	P3	P4	P5
Burst time	5	13	8	4	10
Arrival time	2	3	0	5	1

Answer

(1) Using preemptive short-job first

Gantt chart



W.T. of p1 = 2 -2 = 0

W.T. of p2 = 27 – 3 = 24

W.T. of p3 = 0+ (11-2) = 9

W.T. of p4 = 7 – 5 = 2

W.T. of p5 = 17 – 1 = 16

T.A.T. of P1=7 – 2 = 5

T.A.T. of P2= 40 – 3 = 37

T.A.T. of P3= 17 – 0 = 17

T.A.T. of P4= 11 – 5 = 6

T.A.T. of P5= 27 -1 = 26

A.W.T. = (0+24+9+2+16)/5 = 51/5 = 10.2

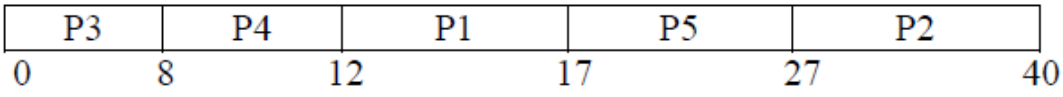
A.T.A.T = (5+37+17+6+26)/5 = 91/5 =18.2

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2) **Using non- preemptive short-job first**

Gantt chart



W.T. of p1 = 12 -2 = 10 T.A.T. of P1= 17 – 2 = 15

W.T. of p2 = 27 – 3 = 24 T.A.T. of P2= 40 – 3 = 37

W.T. of p3 = 0 T.A.T. of P3= 8 – 0 = 8

W.T. of p4 = 8 – 5 = 3 T.A.T. of P4= 12 – 5 = 7

W.T. of p5 = 17 – 1 = 16 T.A.T. of P5= 27 -1 = 26

A.W.T. = (10+24+0+3+16)/5 = 53/5 = 10.6

A.T.A.T = (15+37+8+7+26)/5 = 93/5 = 18.6

Q3: Find the average waiting time and turnaround time for executing the following process using priority scheduling algorithm?

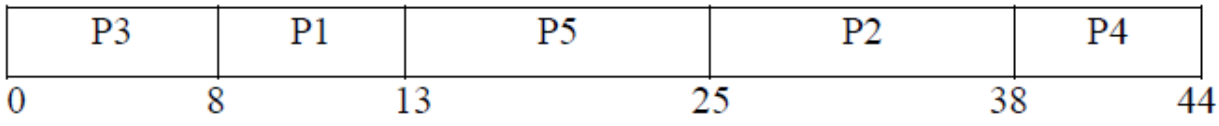
Process	P1	P2	P3	P4	P5
Burst time	5	13	8	6	12
Priority	1	3	0	4	2

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Answer

Gantt chart



W.T. of p1 = 8

W.T. of p2 = 25

W.T. of p3 = 0

W.T. of p4 = 38

W.T. of p5 = 13

T.A.T. of P1=13

T.A.T. of P2= 38

T.A.T. of P3= 8

T.A.T. of P4= 44

T.A.T. of P5= 25

A.W.T. = (8+25+0+38+13)/5 = 84/5 = 16.8

A.T.A.T = (13+38+8+44+25)/5 = 128/5 =25.6

Q4: Find the average waiting time (A.W.T.) and the average turnaround time (A.T.A.T.) for executing the following processes using round-robin algorithm, where time quantum is 5?

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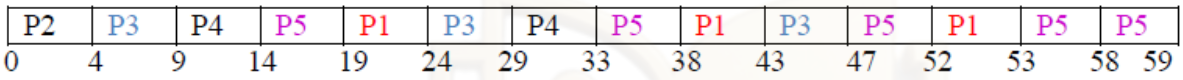
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Process	P1	P2	P3	P4	P5
Burst time	11	4	14	9	21
Arrival time	5	0	0	1	2

Answer

P2= ~~4~~ 0
P3= ~~14~~ ~~9~~ ~~4~~ 0
P4= ~~9~~ ~~4~~ 0
P5=~~21~~ ~~16~~ ~~11~~ ~~6~~ ~~1~~ 0
P1= ~~11~~ ~~6~~ ~~1~~ 0

Gantt chart



W.T. of P1 =(19-5)+(38-24)+(52-43) = 14 + 14+9 = **37**
W.T. of P2 = **0**
W.T. of P3 = (4-0) + (24-9) + (43-29) = 4 + 15 + 14 = **33**
W.T. of P4 = (9-1) + (29-14) = 8 + 15 = **23**
W.T. of P5 = (14-2) + (33-19) + (47-38) + (53-52) = 12+14+9+1=**36**
A.W.T. = (37 + 0+ 33 + 23 + 36) /5 = 129 /5 = 25.8

T.A.T of P1 = (53-5)= **48**
T.A.T of P2 = (4-0) = **4**
T.A.T of P 3= (47 – 0) = **47**
T.A.T of P 4= (33-1) = **32**
T.A.T of P 5= (59-2) = **57**
A.T.A.T. = (48 + 4+ 47 + 32 + 57) /5 = 188/5 = 37.6

Q25: Consider a multilevel feedback queue scheduling (MLFBQ) with three queues q1, q2, and q3. q1 and q2 use round-robin algorithm with time quantum (TQ) = 5, and 4 respectively. q3 use first-come first-service algorithm. Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process?

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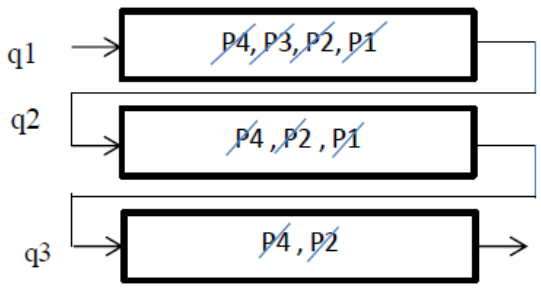
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Processes	P1	P2	P3	P4
Burst time	8	22	4	12

Answer

In MLFBQ scheduling algorithm, the process move between queues. If a process uses too much CPU time, it will be moved to a lower-priority queue.

	q1	q2	q3
P1:	~~8~~	3	-
P2:	22	17	13
P3:	4	-	-
P4:	12	7	3



Gantt chart

P1	P2	P3	P4	P1	P2	P4	P2	P4
0	5	10	14	19	22	26	30	43	46

W.T. of p1 = 0 + (19-5) = 14 T.A.T. of P1=22

W.T. of p2 = 5+ (22-10) + (30-26) = 5+12+4 = 21 T.A.T. of P2= 43

W.T. of p3 = 10 T.A.T. of P3= 14

W.T. of p4 = 14 + (26-19) + (43-30) = 14+7+13= 34 T.A.T. of P4= 46

A.W.T. = (14 + 21 +10 + 34)/4 = 79/4 = 19.75

A.T.A.T = (22 + 43 + 14 + 46) = 125/4 = 31.25

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